BASE CHANGE OF THE RELATIVE SPEC FUNCTOR

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Pullback compatibility. definitional identity. For f = g o f-prime the pullbacks (f-prime)^* of g^* of A and f^* of A agree by functoriality of pullback.

Source reference. Original reference 01LM: sources/stacks-constructions/pinned-original-source.tex, source lines 530–530. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 01LM. Let F be the functor associated to $(S, \mathcal{A})$ above. Let $g : S' \to S$ be a morphism of schemes. Set $\mathcal{A}' = g^*\mathcal{A}$. Let F' be the functor associated to $(S', \mathcal{A}')$ above. Then there is a canonical isomorphism*

$$F' \cong h_{S'} \times_{h_S} F$$

of functors.

Proof. A pair $(f' : T \to S', \varphi' : (f')^*\mathcal{A}' \to \mathcal{O}_T)$ is the same as a pair $(f, \varphi : f^*\mathcal{A} \to \mathcal{O}_T)$ together with a factorization of f as $f = g \circ f'$. Namely with this notation we have $(f')^*\mathcal{A}' = (f')^*g^*\mathcal{A} = f^*\mathcal{A}$. Hence the lemma. □

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.